

serving you a banana daiquiri. You know that the two states are mutually exclusive (you can only be in one place at one time), but exhaustive (there is a 100% probability that you will be in one of them). They are equiprobable, with, in your opinion, 50% probability assigned to each.

You derive great pleasure thinking about your vacation; it motivates you and makes your daily commute more bearable. But the adequate way to visualize yourself, according to rational behavior under uncertainty, is 50% in one of the vacation spots and 50% in the other – what is mathematically called a *linear combination* of the two states. Can your brain handle that? How desirable would it be to have your feet in the Caribbean waters and your head exposed to the Parisian rain? Our brain can properly handle one and only one state at once – unless you have personality troubles of a deeply pathological nature. Now try to imagine an 85%/15% combination. Any luck?

Consider a bet you make with a colleague for the amount of \$1,000, which, in your opinion, is exactly fair. Tomorrow night you will have zero or \$2,000 in your pocket, each with a 50% probability. In purely mathematical terms, the fair value of a bet is the linear combination of the states, here called the *mathematical expectation*, i.e., the probabilities of each payoff multiplied by the dollar values at stake (50% multiplied by 0 and 50% multiplied by \$2,000 = \$1,000). Can you *imagine* (that is visualize, not compute mathematically) the value being \$1,000? We can conjure up one and only *one* state at a given time. Left to our own devices, we are likely to bet in an irrational way, as one of the states would dominate the picture.

Some Architectural Considerations

Time to reveal Nero's secret. It was a black swan. He was then 35. Although pre-war buildings in New York can have a pleasant front, their architecture seen from the back offers a stark contrast by being completely bland. The doctor's examination room had a window overlooking the backyard of one such Upper East side street, and Nero

will always remember how bland that backyard was in comparison with the front, even if he were to live another half century. He will always remember the view of the ugly pink backyard from the leaden window panes, and the medical diploma on the wall that he read a dozen times as he was waiting for the doctor to come into the room (half an eternity, for Nero suspected that something was wrong). The news was then delivered (grave voice), "I have some ... I got the pathology report ... It's ... It is not as bad as it sounds ... It's ... It's cancer". The declaration caused his body to be hit by an electric discharge, running through his back down to his knees. Nero tried to yell "what?" but no sound came out of his mouth. What scared him was not so much the news as the sight of the doctor. Somehow the news reached his body before his mind. There was too much fear in the doctor's eyes and Nero immediately suspected that the news was even worse than what he was being told (it was).

The night of the diagnosis, at the medical library where he sat, drenched wet from walking for hours in the rain without noticing it and making a puddle of water around him (he was yelled at by an attendant but could not concentrate on what she was saying so she shrugged her shoulders and walked away); later he read the sentence "72% 5-year actuarially adjusted survival rate". It meant that 72 people out of a hundred make it. It takes between three and five years for the body without clinical manifestations of the disease for the patient to be pronounced cured (closer to three at his age). He then felt in his guts quite certain that he was going to make it.

Now the reader might wonder about the mathematical difference between a 28% chance of death and a 72% chance of survival over the next five years. Clearly, there is none, but we are not made for mathematics. In Nero's mind a 28% chance of death meant the image of himself dead, and thoughts of the cumbersome details of his funeral. A 72% chance of survival put him in a cheerful mood; his mind was planning the result of a cured Nero skiing in the Alps. At no point during his ordeal did Nero think of himself as 72% alive and 28% dead.